HW4

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Contents

```
library(RMariaDB)
```

```
## Warning: package 'RMariaDB' was built under R version 4.4.3
```

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3

## Warning: package 'ggplot2' was built under R version 4.4.3

## Warning: package 'tibble' was built under R version 4.4.3

## Warning: package 'tidyr' was built under R version 4.4.3

## Warning: package 'readr' was built under R version 4.4.3

## Warning: package 'purrr' was built under R version 4.4.3

## Warning: package 'dplyr' was built under R version 4.4.3

## Warning: package 'stringr' was built under R version 4.4.3

## Warning: package 'forcats' was built under R version 4.4.3

## Warning: package 'lubridate' was built under R version 4.4.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(mdsr)
```

```
## Warning: package 'mdsr' was built under R version 4.4.3
```

Q2

```r
con <- dbConnect(
  MariaDB(), host = "scidb.smith.edu",
  user = "waiuser", password = "smith_waiDB",
  dbname = "wai"
)

Measurements <- tbl(con, "Measurements")

dbGetQuery(con, "SELECT COUNT(*) AS n_rows FROM Measurements;")
```

```
##    n_rows
## 1 5052304
```

```r
dbDisconnect(con)
```

Q3

```r
con <- dbConnect_scidb("airlines")

dbGetQuery(con, "
  SELECT year, COUNT(*) AS n_flights
  FROM flights
  GROUP BY year
  ORDER BY year;
")
```

```
##   year n_flights
## 1 2013   6369482
## 2 2014   5819811
## 3 2015   5819079
```

Q4

```r
# 4.1
dbGetQuery(con, "
           SELECT COUNT(*) AS n_flights
           FROM flights
           WHERE year = 2015
            AND month = 5
            AND day = 14
            AND dest = 'DFW';")
```

```
##   n_flights
## 1       737
```

```
# 4.2
dbGetQuery(con, "
  SELECT dest, COUNT(*) AS n_flights
  FROM flights
  WHERE year = 2015 AND origin = 'ORD'
  GROUP BY dest
  ORDER BY n_flights DESC
  LIMIT 10;
")
```

```
##    dest n_flights
## 1   LGA     10492
## 2   LAX      8720
## 3   DFW      8384
## 4   SFO      8156
## 5   BOS      7240
## 6   ATL      7104
## 7   MSP      6955
## 8   DCA      6495
## 9   DEN      6120
## 10  MKE      5274
```

```
# 4.3
dbGetQuery(con, "
  SELECT dest, AVG(arr_delay) AS avg_arr_delay, COUNT(*) AS n_flights
  FROM flights
  WHERE year = 2015
  GROUP BY dest
  ORDER BY avg_arr_delay DESC
  LIMIT 10;
")
```

```
##    dest avg_arr_delay n_flights
## 1   STC       21.6220        82
## 2   ILG       21.3163        98
## 3   OTH       17.0868       311
## 4   TTN       15.7600      3113
## 5   PBG       15.3559       295
## 6   GUM       14.3671       365
## 7   ASE       14.3334      3620
## 8   MQT       14.1420       331
## 9   BPT       12.9792      1060
## 10  BQN       11.6724      1441
```

```
# 4.4
dbGetQuery(con, "
  SELECT COUNT(*) AS n_flights
  FROM flights
  WHERE year = 2015
    AND (origin = 'BDL' OR dest = 'BDL');
")
```

```
##   n_flights
## 1     41025
```

```
# 4.5
dbGetQuery(con, "
  SELECT c.name AS airline, f.carrier, f.flight, f.origin, f.dest
  FROM flights AS f
  JOIN carriers AS c ON f.carrier = c.carrier
  WHERE f.year = 2015 AND f.month = 9 AND f.day = 26
    AND ((f.origin = 'LAX' AND f.dest = 'JFK')
      OR (f.origin = 'JFK' AND f.dest = 'LAX'))
  ORDER BY f.carrier, f.flight;
")
```

```
##                   airline carrier flight origin dest
## 1  American Airlines Inc.      AA      2    LAX  JFK
## 2  American Airlines Inc.      AA      3    JFK  LAX
## 3  American Airlines Inc.      AA      4    LAX  JFK
## 4  American Airlines Inc.      AA     19    JFK  LAX
## 5  American Airlines Inc.      AA     21    JFK  LAX
## 6  American Airlines Inc.      AA     22    LAX  JFK
## 7  American Airlines Inc.      AA     30    LAX  JFK
## 8  American Airlines Inc.      AA     32    LAX  JFK
## 9  American Airlines Inc.      AA     33    JFK  LAX
## 10 American Airlines Inc.      AA     34    LAX  JFK
## 11 American Airlines Inc.      AA    117    JFK  LAX
## 12 American Airlines Inc.      AA    118    LAX  JFK
## 13 American Airlines Inc.      AA    180    LAX  JFK
## 14 American Airlines Inc.      AA    255    JFK  LAX
## 15 American Airlines Inc.      AA    293    JFK  LAX
## 16        JetBlue Airways      B6     23    JFK  LAX
## 17        JetBlue Airways      B6     24    LAX  JFK
## 18        JetBlue Airways      B6    123    JFK  LAX
## 19        JetBlue Airways      B6    124    LAX  JFK
## 20        JetBlue Airways      B6    223    JFK  LAX
## 21        JetBlue Airways      B6    224    LAX  JFK
## 22        JetBlue Airways      B6    323    JFK  LAX
## 23        JetBlue Airways      B6    324    LAX  JFK
## 24        JetBlue Airways      B6    423    JFK  LAX
## 25        JetBlue Airways      B6    424    LAX  JFK
## 26        JetBlue Airways      B6    523    JFK  LAX
## 27        JetBlue Airways      B6    524    LAX  JFK
## 28        JetBlue Airways      B6    623    JFK  LAX
## 29        JetBlue Airways      B6    624    LAX  JFK
## 30   Delta Air Lines Inc.      DL    412    LAX  JFK
## 31   Delta Air Lines Inc.      DL    420    JFK  LAX
## 32   Delta Air Lines Inc.      DL    423    JFK  LAX
## 33   Delta Air Lines Inc.      DL    447    JFK  LAX
## 34   Delta Air Lines Inc.      DL    464    JFK  LAX
## 35   Delta Air Lines Inc.      DL    472    JFK  LAX
## 36   Delta Air Lines Inc.      DL    476    LAX  JFK
## 37   Delta Air Lines Inc.      DL    477    JFK  LAX
## 38   Delta Air Lines Inc.      DL    920    LAX  JFK
## 39   Delta Air Lines Inc.      DL   1162    LAX  JFK
```

```
## 40   Delta Air Lines Inc.      DL   1262    LAX  JFK
## 41   Delta Air Lines Inc.      DL   1908    LAX  JFK
## 42  United Air Lines Inc.      UA    275    LAX  JFK
## 43  United Air Lines Inc.      UA    441    JFK  LAX
## 44  United Air Lines Inc.      UA    535    JFK  LAX
## 45  United Air Lines Inc.      UA    703    JFK  LAX
## 46  United Air Lines Inc.      UA    779    LAX  JFK
## 47  United Air Lines Inc.      UA    841    JFK  LAX
## 48  United Air Lines Inc.      UA    912    LAX  JFK
## 49  United Air Lines Inc.      UA   1752    LAX  JFK
## 50  United Air Lines Inc.      UA   1985    JFK  LAX
## 51         Virgin America      VX    399    JFK  LAX
## 52         Virgin America      VX    404    LAX  JFK
## 53         Virgin America      VX    406    LAX  JFK
## 54         Virgin America      VX    407    JFK  LAX
## 55         Virgin America      VX    411    JFK  LAX
## 56         Virgin America      VX    412    LAX  JFK
## 57         Virgin America      VX    413    JFK  LAX
## 58         Virgin America      VX    415    JFK  LAX
## 59         Virgin America      VX    416    LAX  JFK
## 60         Virgin America      VX    420    LAX  JFK
```

```
dbDisconnect(con)
```